

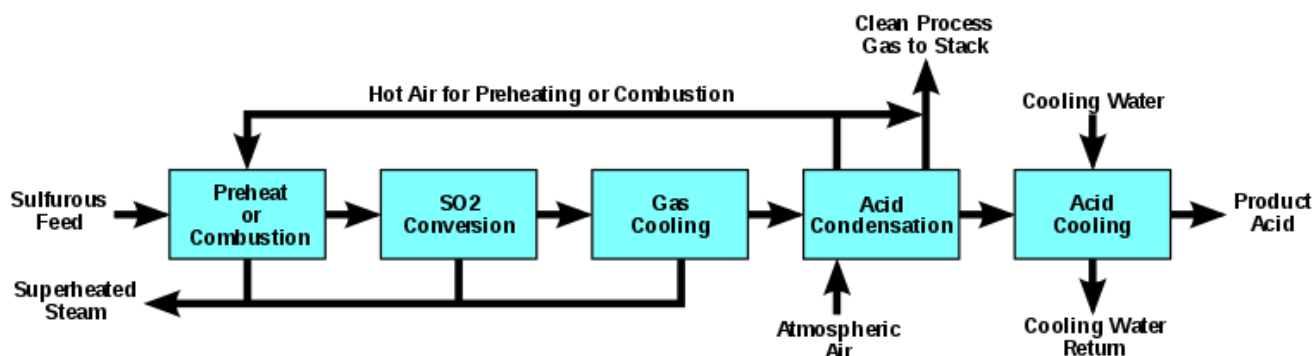
Wet sulfuric acid process

The **wet sulfuric acid process** (**WSA process**) is a gas [desulfurization](#) process. After Danish company [Haldor Topsoe](#) introduced this technology in 1987, it has been recognized as a process for recovering sulfur from various process gases in the form of commercial quality sulfuric acid (H_2SO_4) with the simultaneous production of high-pressure steam. The WSA process can be applied in all industries where sulfur removal presents an issue.

The wet catalysis process is used for processing sulfur-containing streams, such as:^[1]

- H_2S gas from e.g. [amine gas treating unit](#)
- Off-gas from [sour water stripper \(SWS\)](#) gas
- Off-gas from [Rectisol](#)
- Spent acid from an [alkylation unit](#)
- [Claus process](#) tail gas
- Heavy residue or [petcoke](#)-fired utility boiler off-gas
- Boiler flue gases from various processes [SNOX flue gas desulfurization](#)
- Metallurgical process gas
- Production of [sulfuric acid](#)

The process



The main reactions in the WSA process

- Combustion: $2 \text{H}_2\text{S} + 3 \text{O}_2 \rightleftharpoons 2 \text{H}_2\text{O} + 2 \text{SO}_2$ (-1036 kJ/mol)
- Oxidation: $2 \text{SO}_2 + \text{O}_2 \rightleftharpoons 2 \text{SO}_3$ (-198 kJ/mol) [in the presence of a [vanadium \(V\) oxide catalyst](#)]
- Hydration: $\text{SO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{SO}_4 (\text{g})$ (-101 kJ/mol)
- Condensation: $\text{H}_2\text{SO}_4 (\text{g}) \rightleftharpoons \text{H}_2\text{SO}_4 (\text{l})$ (-90 kJ/mol)

The energy released by the above-mentioned reactions is used for steam production. Approximately 2–3 tons of high-pressure steam are produced per ton of acid.

Industrial applications

Industries where WSA process plants are installed:

- Refinery and [petrochemical industry](#)
- [Metallurgy industry](#)
- [Coal-based industry](#) (coking and gasification)

- Power industry
- Viscose industry
- Sulfuric acid industry

WSA for gasifiers

The acid gas coming from a Rectisol-, Selexol-, amine gas treating or similar unit installed after the gasifier contains H₂S, COS and hydrocarbons in addition to CO₂. These gases were previously vented to the atmosphere, but now the acid gas requires purification in order not to affect the environment with SO₂ emission. The WSA process provides a high sulfur recovery and recovers heat for steam production. The heat recovery rate is high, and the cooling water consumption is low, which saves resources.^[2]

Examples of WSA process for gasification

Example 1:

- Feed-gas flow: 14,000 Nm³/h
- Composition [vol %]: 5.8% H₂S, 1.2% COS, 9.7% HC and 77.4% CO₂
- SO_x concentration [vol %]: 1.58%
- H₂SO₄ production: 106 MTPD
- Steam production: 53 ton/h
- Cooling water consumption: 8 m³/ton acid (delta T = 10 °C)
- Fuel consumption: 1,000 Nm³/h (LHV = 2,821 kcal/Nm³)

Example 2: A sulfur plant in China will be built in connection with an ammonia plant, producing 500 kilotons/year of ammonia for fertilizer production ^[3]

Spent acid regeneration and production of sulfuric acid

The WSA process can also be used for the production of sulfuric acid from sulfur burning or regeneration of the spent acid from e.g., alkylation plants. Wet catalysis processes differ from other contact sulfuric acid processes in that the feed gas contains excess moisture when it comes into contact with the catalyst. The sulfur trioxide formed by catalytic oxidation of the sulfur dioxide reacts instantly with the moisture to produce sulfuric acid in the vapor phase to an extent determined by the temperature. Liquid acid is subsequently formed by condensation of the sulfuric acid vapor and not by the absorption of the sulfur trioxide in concentrated sulfuric acid, as in contact processes based on dry gases.

The concentration of the product acid depends on the H₂O:SO₃ ratio in the catalytically converted gases and on the condensation temperature.^{[4][5]}

The combustion gases are cooled to the converter inlet temperature of about 420–440 °C. Processing these wet gases in a conventional cold-gas contact process (DCDA) plant would necessitate cooling and drying of the gas to remove all moisture. Therefore, the WSA process is, in most cases, a more cost-efficient way of producing sulfuric acid.

About 80% to 85% of the world's sulfur production is used to manufacture sulfuric acid. 50% of the world's sulfuric acid production is used in fertilizer production, mainly to convert phosphates to water-soluble forms. according to the Fertilizer Manual published jointly by the United Nations Industrial Development Organization (UNIDO) and the International Fertilizer Development Center ^[6]

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